

RC CAR SIDEWALK MODEL PLAN

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This document describes a practical plan for building a sidewalk-detection model for the RC car using real data from the car's own camera setup.

1. OBJECTIVE

The goal is to replace the current fragile classical sidewalk-edge detector with a trained segmentation model that can identify the drivable sidewalk region more reliably.

The model should support:

- sidewalk following
- turn handling
- shade robustness
- road avoidance
- grass avoidance
- obstacle-aware path interpretation

The output of the model should be a sidewalk mask, not just a box.

That mask can then be converted into:

- a sidewalk region
- a centerline
- a target heading
- a confidence score

2. RECOMMENDED MODEL TYPE

Use segmentation, not bounding-box object detection.

Reason:

- steering needs the shape of the drivable region
- a box around "sidewalk" is not enough to steer accurately
- segmentation gives a mask that can be converted into boundaries and a centerline

3. RECOMMENDED CLASSES

3.1 BEST VERSION 1 CLASS SET

Start with these four classes:

- `sidewalk`
- `grass`
- `road`
- `obstacle`

3.2 SMALLER VERSION 1 ALTERNATIVE

If you need a simpler first pass, use:

- `sidewalk`
- `not\_sidewalk`

This is easier to label, but gives less control information for later planning.

The recommended choice is still the 4-class version.

4. HOW MANY IMAGES

4.1 MINIMUM USABLE START

- `500` labeled images

This is enough to test the pipeline, but not enough for strong robustness.

4.2 BETTER FIRST REAL DATASET

- `1000` to `2000` labeled images

4.3 RECOMMENDED TARGET

Aim first for:

- `1200` total labeled images

Recommended split:

- `800` training
- `200` validation
- `200` test

4.4 STRONGER LATER VERSION

After the first round, expand toward:

- `2500+` total images

5. WHAT TO COLLECT

Do not use random internet sidewalk photos.

Collect data from your actual RC car camera, with the actual mounting height, tilt, lens, and field of view.

The dataset should include:

- straight sidewalks
- gentle turns
- sharp turns
- shaded sidewalks
- bright sun
- mixed lighting
- driveway cuts
- sidewalk next to grass
- sidewalk next to road
- narrow sidewalks
- wide sidewalks
- partial obstacles
- pedestrians
- cracks and seams
- leaves and debris

Also vary:

- time of day
- sun angle
- left/right path offsets
- slight camera motion and vibration

6. RECOMMENDED DISTRIBUTION FOR 1200 IMAGES

A balanced first dataset could be:

- `300` straight clean sidewalk images
- `200` shade or dappled-light images
- `150` gentle turns
- `100` sharper turns
- `100` driveway-cut cases
- `100` sidewalk-road boundary cases

- `100` sidewalk-grass boundary cases
- `100` partial obstacles or pedestrians
- `50` failure-case images
- `100` mixed miscellaneous images

7. BEST COLLECTION METHOD

Do not manually take hundreds of individual still photos.

Instead:

1. record video from the actual RC car camera
2. drive or walk through representative sidewalk routes
3. extract frames every `0.5` to `1.0` seconds
4. remove near-duplicate frames
5. keep difficult and varied scenes

This is much faster and produces better training data.

8. LABELING RULES

Use Roboflow to label the scene according to driving behavior.

8.1 SIDEWALK

Label the true drivable sidewalk surface as `sidewalk`.

8.2 GRASS

Label grass, landscaping, and lawn regions as `grass`.

8.3 ROAD

Label asphalt or road surfaces as `road`.

8.4 OBSTACLE

Label things the car must not drive through as `obstacle`, such as:

- pedestrians
- cones
- bins
- poles
- parked-car intrusion
- large debris

9. IMPORTANT LABELING PRINCIPLE

Label according to control behavior, not just appearance.

That means:

- if the car may drive there, it should become `sidewalk`
- if the car must never drive there, it should become `road`, `grass`, or `obstacle`

9.1 SHADOWS

Shadows should not be their own class.

Instead:

- label the real surface underneath the shadow

9.2 DRIVEWAY CUTS

Because your rule is to stop at driveway cuts instead of treating them like normal sidewalk continuation, these scenes need especially careful and consistent labeling.

10. TRAINING PLAN

Use an iterative process.

10.1 FIRST ROUND

1. collect `300` images
2. label them
3. train a first segmentation model
4. check whether the pipeline works at all

10.2 SECOND ROUND

1. review failure cases
2. collect `400` more images focused on failures
3. relabel and retrain

10.3 THIRD ROUND

1. collect another `500+` images for robustness
2. retrain
3. evaluate on a held-out test set

This is better than collecting all `1200` blindly before learning where the model fails.

11. EVALUATION CRITERIA

Do not rely only on training metrics.

The real questions are:

- does the sidewalk mask cover the true sidewalk?
- does it avoid road and grass?
- does it survive shade?
- does it remain stable through turns?
- does it behave correctly at driveway cuts?
- does it remain usable when obstacles appear?

12. WHAT GOOD ENOUGH LOOKS LIKE

Before using it for steering, the model should:

- keep the sidewalk mask connected
- produce a stable centerline
- avoid treating road as sidewalk
- avoid treating grass as sidewalk
- remain reasonably stable in shade
- remain usable at turns

13. RUNTIME INTEGRATION PLAN

Once trained, the camera pipeline should become:

1. capture camera frame
2. run segmentation model
3. produce sidewalk mask
4. choose the bottom-center connected sidewalk region
5. compute centerline and target heading
6. compute confidence
7. fuse with LiDAR obstacle geometry
8. send one steering target to the servo

14. FINAL RECOMMENDATION

For version 1, the recommended path is:

- use Roboflow
- train a segmentation model
- start with `1200` labeled images
- use the actual RC car camera setup
- include lots of shade, turns, and driveway-cut scenes
- expand later using hard failure cases

This is a much more promising path than continuing to patch the current edge-based classical detector.